

Methodology and Data

Methodology

Our framework is organized around alternative specifications of the New Keynesian Phillips Curve with hybrid expectational structure — combining forward-looking and backward-looking inflation terms — while varying the measure of real economic slack.

$$\pi_t = \gamma_f E_t\{\pi_{t+1}\} + \gamma_b \pi_{t-1} + \kappa \hat{s}_t \quad (1)$$

where π_t is the inflation rate and $\hat{s}_t$ is a measure of real economic slack. The coefficient γ_f governs the weight agents place on forward-looking inflation expectations, while γ_b captures the degree of backward-looking behavior or intrinsic inflation persistence, and κ is the slope of the Phillips curve with respect to the chosen slack measure. Following the literature, $\hat{s}_t$ can be operationalised as the real marginal cost gap $\widehat{mc}_t$ (Galí and Gertler, 1999), the output gap $\hat{y}_t$ (Galí et al., 2001), or the unemployment gap $\hat{u}_t$ (Blanchard and Galí, 2010; Hazell et al., 2022), with the sign of κ depending on the chosen measure — positive for output- and cost-based proxies, negative for unemployment.

To estimate the Equation (1), we employ a two-way fixed effects panel framework using EMU member states and quarterly data. The baseline specification takes the following form:

$$\pi_{i,t} = \alpha_i + \gamma_t + \beta_1 s_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t} \quad (2)$$

where $\pi_{i,t}$ denotes inflation in country i at time t , measured alternatively as headline, core, or non-tradable inflation to assess the robustness and fit of the slack-inflation relationship across different price indices. α_i represents country fixed effects, capturing time-invariant cross-country heterogeneity. γ_t denotes time fixed effects, absorbing common macroeconomic shocks shared across all countries in a given period, including EMU-wide inflation expectations, changes in potential unemployment (GDP) and common monetary policy stances. $s_{i,t}$ is the measure of economic slack, proxied alternatively by the unemployment rate, the real GDP, and the labour share of income, each capturing a different dimension of demand pressure and factor market tightness. The composition of the control vector $X_{i,t}$ varies across

specifications. Where included, the control set comprises: lagged inflation $\pi_{i,t-1}$, which accounts for inflation persistence and accommodates the accelerationist interpretation of the Phillips curve — whereby past inflation shapes current price-setting behaviour; imported price index $p_{i,t}^m$, which controls for external cost-push pressures stemming from country-specific import prices and exchange rate movements; and the relative price of non-tradables $p_{i,t}^{nt}$, which captures domestic supply-side cost pressures and sectoral price dynamics that may influence inflation independently of demand conditions. Finally, $\varepsilon_{i,t}$ denotes the idiosyncratic error term.

The two-way fixed effects structure embedded in the baseline specification serves a dual identification purpose, and its use in the context of Phillips curve estimation is well established in the literature. Country fixed effects, α_i , absorb all time-invariant cross-country heterogeneity, including structural differences in the natural rate of unemployment, demographic and industrial composition, the generosity of social insurance systems as well as constant productivity growth differentials, and time-invariant differences in inflation expectations across member states (Aaronson and Sullivan, 2000; McLeay and Tenreyro, 2019; Fitzgerald et al., 2024).

Time fixed effects, γ_t , play an equally important role by absorbing all shocks that are common across countries in a given period. As Furlanetto and Lepetit (2024) note, this eliminates the endogeneity bias induced by the response of monetary policy to cost-push factors, a point emphasised by McLeay and Tenreyro (2019), and additionally controls for time variation in aggregate inflation expectations and movements in aggregate potential output. In the EMU context, this is particularly valuable, as the single monetary policy stance of the ECB constitutes precisely such a common time-varying factor. By absorbing it through time fixed effects, the specification isolates the within-country, within-period variation in slack that is orthogonal to the common policy cycle, thereby providing a cleaner basis for identifying the slope of the Phillips curve (Babb and Detmeister, 2017; Vaona, 2007). The ability to control for EMU wide inflation expectations through time fixed effects is, as Furlanetto and Lepetit (2024) emphasize, among the principal advantages of the panel approach to Phillips curve estimation.

While long-run inflation expectations should be independent of the business cycle and are

primarily a function of the credibility of monetary policy — and can therefore be controlled for by time fixed effects — short-run expectations may differ across member states due to differences in economic circumstances (natural rate) or cultural factors. To the extent that such differences are time-invariant, they will be captured by country fixed effects; where they reflect time-varying idiosyncratic dynamics, such as shocks to natural rate, volatility or gradual convergence toward the euro area norm, they will be absorbed by the error term (Hazell et al., 2022).

Equation (3) formalises the correspondence between the theoretical NKPC in Equation (1) and its panel econometric counterpart in Equation (2), making explicit the identifying assumption underlying the two-way fixed effects approach: the forward-looking expectation term $\gamma_f E_t[\pi_{it+1}]$ is proxied with time fixed effects λ_t absorbing common long-run expectations shaped by the credibility of monetary policy, country fixed effects μ_i capturing time-invariant cross-country heterogeneity in short-run expectation formation arising from differences in economic structure or cultural factors as well as differences in the natural rate of unemployment, and the idiosyncratic error term ε_{it} accounting for residual time-varying dynamics such as gradual convergence toward the euro area inflation norm and/or shocks to the natural rate of unemployment.

$$\underbrace{\pi_{it}}_{\text{inflation}} = \underbrace{\gamma_b \pi_{it-1}}_{\text{backward-looking expectations}} + \underbrace{\kappa s_{it}}_{\text{unemployment}} + \underbrace{\lambda_t}_{\substack{E[\pi^{LR}] \\ \text{monetary credibility}}} + \underbrace{\mu_i}_{\substack{E[\pi_i^{SR}] \\ \text{cultural/structural} \\ \text{heterogeneity} \\ s_i^n}} + \underbrace{\varepsilon_{it}}_{\substack{\text{idiosyncratic} \\ \text{convergence dynamics,} \\ \Delta s_{it}^n}} \quad (3)$$

In the specification that includes lagged inflation as a control, we also examine how accounting for inflation persistence affects the estimated fixed-effect coefficients and error terms — serving as a test of whether backward-looking expectations can explain part of the country-specific short-run variation in the basic specification.

The model is estimated using four complementary approaches. The baseline specification relies on the `ivreghdfe` command, which combines the fixed-effects absorbing algorithm of `reghdfe` (Correia and Constantine, 2014; Correia, 2017) with two-stage least squares (Baum et al., 2002). To address the potential endogeneity of the slack variable arising from

simultaneity between inflation and economic activity, the slack variable is instrumented with its own two lags, exploiting the serial persistence of cyclical conditions.

$$\pi_{i,t} = \alpha_i + \gamma_t + \beta_1 \hat{s}_{i,t} + \beta_2 X_{i,t} + \varepsilon_{i,t}, \quad \hat{s}_{i,t} = f(s_{i,t-1}, s_{i,t-2}) \quad (4)$$

where $\hat{s}_{i,t}$ denotes the fitted values from the first stage in which $s_{i,t}$ is projected onto $s_{i,t-1}$ and $s_{i,t-2}$, alongside the full set of country and time fixed effects.

As a robustness check, we re-estimate the Equation (2) using the `reghdfe` command, which efficiently absorbs high-dimensional fixed effects without instrumentation, allowing us to assess the sensitivity of the results to the instrumentation strategy.

To account for the dynamic nature of the model and the Nickell bias introduced by the inclusion of a lagged dependent variable in a fixed effects panel, we additionally estimate the model using the System GMM estimator implemented in `xtabond2` (Roodman, 2009), which instruments the model with Arellano-Bond GMM-style instruments for endogenous variables and with external instruments for time dummies (Arellano and Bover, 1995; Blundell and Bond, 1998).

Finally, we employ the `ts2sls` estimator (Chodorow-Reich and Wieland, 2020) following Hazell et al. (2022) as a further robustness check on the IV results. Standard errors are clustered at the country level throughout to account for within-country serial correlation and heteroskedasticity.

Data

For the analysis of the heterogeneous Phillips curve, we use variables from 4 Eurostat databases (Eurostat, 2022a,b, 2025a,b) and real exchange rate data from ECB database (European Central Bank, 2026).

The analysis covers 20 member states of the Eurozone, excluding Bulgaria, which joined in 2026. To obtain the variables needed to measure the output gap and construct the GDP deflator, we use the `namq_10_gdp` database (Eurostat, 2025b). This enables the use of quarterly GDP at market prices alongside real GDP, with 2020 base year. These series are then used to construct the GDP deflator, which captures movements in domestic prices. Addi-

tionally, compensation of employees (D1) is used to construct the labour income share in GDP, which is calculated as the ratio of compensation of employees to nominal GDP.

Since the heterogeneous Phillips curve requires different components of inflation dynamics, we also include various measures of consumer price indices at a monthly frequency, allowing us to capture the influence of imported goods in addition to the GDP deflator. For this purpose, the Eurostat database `prc_hicp_midx` (Eurostat, 2022a) is utilized. As a measure of aggregate inflation, we use the Harmonised Index of Consumer Prices (HICP), with 2015 as the base year. For the individual components of inflation, we use measures of core inflation and 41 sectorally disaggregated price indices at the COICOP 3-digit level. This enables us to analyze the heterogeneity of inflation dynamics across major consumption categories, including food, energy and services.

For the economic slack variables, in addition to the variables obtained from the `namq_10_gdp` database (Eurostat, 2025b), two additional databases are used. The first database, `une_rt_m` (Eurostat, 2025a), contains data on various monthly unemployment indicators, including seasonally adjusted series expressed as a percentage of the active population, as well as corresponding levels in thousands of persons. The second database, `lfsi_emp_q` (Eurostat, 2022b), contains quarterly indicators of employment such as employment and activity rates (ratios to working age population). This will enable us to combine multiple indicators of the economic slack with different measures of inflation, thereby ensuring a robust estimation of the Phillips curve. Since the databases used have different frequencies (monthly and quarterly), it is necessary to harmonize them by aggregating monthly variables to the quarterly level using last month in the quarter collapsing.

In order to create a proxy for import prices we use ECB data on real exchange rates and remove influence of domestic prices from them in order to get trade weighted import price index denominated in euros vis-à-vis 17 major trade partners of each country (European Central Bank, 2026; Schmitz et al., 2024).

The raw series corresponding to the data sources described above are presented in Figures ??, ??, and ?. Figure ?? visualizes the inflation measures constructed from Eurostat `prc_hicp_midx` database, including headline inflation, core inflation, and services inflation (Eurostat, 2022a). Figure ?? reports the unemployment rate from Eurostat `une_rt_m`

database and the labour share constructed from compensation of employees and nominal GDP in `namq_10_gdp` database (Eurostat, 2025a,b). Figure ?? presents real GDP, and the relative price of non-tradables, combining Eurostat `namq_10_gdp` and `prc_hicp_midx` databases with ECB data used to construct the import price proxy (Eurostat, 2025b, 2022a; European Central Bank, 2026; Schmitz et al., 2024).

Descriptive statistics are reported in Table ?. The sample covers 20 EMU member states over roughly 84-91 periods depending on the variable, yielding between 1,276 and 1,815 total observations, with the employment rate having the most limited coverage due to data availability constraints. Inflation measures exhibit substantial within-country variation relative to between-country dispersion, with headline inflation posting an overall standard deviation of 2.53 and ranging from -2.9 to 21.6 percentage points, reflecting the sharp disinflationary and inflationary episodes captured across the sample period. The slack variables display a similar pattern: unemployment varies considerably both across and within countries (between s.d. of 2.83, within s.d. of 2.90), while import prices exhibit the largest overall dispersion of all regressors (s.d. of 19.6), pointing to the potential importance of external price pressures.

Corelation matrix is in table ?. The three inflation measures are strongly mutually correlated (0.73-0.90), and all carry the expected negative correlation with unemployment, which is strongest for services inflation (-0.378), consistent with the more labor-intensive nature of the services sector. Correlation with GDP is also negative while labor share is possitively correlated as theory predicts. Import prices correlate moderately with GDP size (0.414), reflecting the greater openness of smaller economies.

Lastly, Figures ? and ? show the availability of observations for panel estimation. Where Figure ? shows how many observations are available for panel estimation if we exclude the employment rate. For countries that adopted Euro early, the availability of dataset is longest. The only exception are few early years, when data is generally unavailable. As the EMU enlarged, additional countries enter the sample at the time of enlargement. In comparison, Figure ? shows how inclusion of employment rate into analysis shortens the panel, thereby decreasing the number of observations. This is because the employment rate is generally available only from a first quarter of 2009, with France being the main exception. As a result the number of EMU enlargement episodes covered by the given specification is

reduced and omitted from the results section.¹

¹Results are available per request.

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